

WEEKLY REPORT

DENTOBOT Step 6 Engineering Workflow Renovation

29 September 2026

The project has moved from repeated planner-first troubleshooting to a five-DOF, simulation-only engineering workbench that lets the robotics engineer inspect valid and invalid robot states before committing to a route.

SOURCE	Major Step 6 renovation gates are implemented and host-verified.
HEADED GUI	Several bounded simulation slices passed; the final case-bearing campaign remains incomplete.
OPERATOR	Tarun's normal-window robotics and usability verdict is pending.
RUNTIME	Halted by operator after the prepared r14 retry was blocked before launch.

Reporting window: 22-28 September 2026

Pre-context: project state through 21 September 2026

Checkout: feature/step6-workflow-renovation-20260925 | HEAD 84234a6 | dirty changes preserved

1. Executive brief

What is ready, what is not, and why the direction changed

Step 6 is no longer only a planner launch sequence. It is now becoming an engineering environment for understanding the robot: joint-level exploration, Cartesian TCP probing, detached Base and Task Home review, authoritative guarded acceptance, invalid-state diagnostics, ordered evidence, and strict separation between exploratory state and preview authority.

This direction came from repeated evidence that planning failures were not purely algorithmic. Base pose, Task Home, five-DOF reachability, drill-axis continuity, joint limits, collision-scene attribution, insertion envelope and physical robot geometry all influenced the result. Automating more planner retries could produce more logs without giving the robotics engineer the geometric understanding needed to decide whether the task is physically feasible.

5 DOF Current arm problem: J1-J5 only	31 Scene objects acknowledged in bounded runs	394 Indexed five-DOF host selection passed	79 Latest jog + runner host checks passed
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Report map

- 2. Position entering the week - planner history and last known blockers
- 3. Reasoning and planning pivot - why the manual workbench became P0
- 4. Current week's progress - source, UI, runtime and evidence
- 5. Multi-planner comparison workflow and diagnostic findings
- 6. Diagnostic run summaries and screenshot evidence
- 7. Stable, verified and presentable now
- 8. Failures, limitations and evidence boundaries
- 9. Next week - pending plan gates and immediate sequence
- 10. Verification ledger and evidence index

Meeting headline: the architecture and diagnostic capability are substantially improved; the remaining risk is representative case-bearing runtime acceptance, not absence of implementation.

2. Position entering the week

Progress through 21 September and the blockers carried into this reporting window

Date	State carried forward
Sep 19	Ordinary GUI reached Step 6.5 on FDI31, but all 13 planner legs stopped at Home-to-PreEntry P1. P2 and P3 never ran.
Sep 20-21	Diagnosis separated IK/reachability, Home connectivity, collision guard and planner search. A fixed drill frame and strict independent guard remained mandatory.
Sep 22	A housing-off diagnostic passed P1 with 150 waypoints. A separately altered Base/Home/mouth-opening setup reportedly completed a housing-on chain, but its exact state was not saved for independent reproduction.
Sep 22	The Step 5B/5C two-target setup and fresh-process reopen path improved, but canonical planner acceptance and normal-window branch switching remained pending.
Entering week	RRTConnect was the default. RRT and optional RRT* were being added for controlled comparison; approximate IK stayed disabled and collision/phase guards stayed fail-closed.

Last known technical blockers

- Home-to-PreEntry feasibility depended on exact Base, Home, task, scene and robot-profile identity.
- Stage 3 could fail on target-tooth to spindle/housing contact even when earlier stages passed.
- The six-joint historical model mixed the external spindle with the five-axis arm problem.
- Planner comparison was expensive because repeated waypoint guard evaluation dominated runtime.
- Offline collision screenshots could be misleading if their transforms and exact evaluated state did not match the native scene.
- Saved cases could reopen with stale geometry or resource fingerprints; strict validation correctly blocked silent repair.

The unresolved question was broader than 'which planner works?': can the five-axis robot, at a reviewed mount and Home, realize the complete drill-axis task without forbidden contact?

3. Reasoning and planning pivot

From automated planner retries to engineer-led state and motion exploration

The operator clarified the core development need: robotics judgment must enter before repeated planner trial-and-error. The engineer needs to move the simulated robot, see why a state is valid or invalid, inspect TCP and drill-axis behavior, and determine whether a failure comes from planning, geometry, Base placement, Home, or hard design constraints.

Planning decisions adopted

Decision	Practical meaning
Renovation first	Finish the Step 6 engineering workflow before case-specific FDI11/FDI21/FDI31 planner solving.
Five-DOF truth	Remove J6 from the arm state, IK, Home, limits and guard problem. Keep the drill as a separate future speed-controlled device.
State separation	Draft, accepted, monitored and historical states remain distinct. Only an exact authoritative guard acknowledgement may advance accepted simulated state.
Manual authority	Manual, partial and reopened paths are diagnostic only. They never authorize a route or preview.
Two-area UI	Planning & Diagnostics owns exploration and preflight; Preview & Control consumes only a fresh complete guard-valid chain.
Evidence first	Every headed campaign records itemized JSON, state-matched screenshots, logs, video, hashes and owned-process cleanup.

The resulting engineering loop

1. Edit draft	2. Inspect validity	3. Guard request	4. Accept or retain evidence
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Accepted requests mirror into the accepted simulation only after exact request/session/policy/state/scene correlation. Rejected requests keep the draft and reasons. Unknown or stale replies keep the last confirmed accepted state and latch reconciliation. Historical imports are visible but inert.

4. Current week's full progress

22-28 September: implementation, diagnostics, verification and runtime evidence

When	Workstream	Result
22-23 Sep	Planner policy and comparison	RRTConnect/RRT/RRT* chooser, requested/effective identity, persistent diagnostics, frozen comparison identity and per-row checkpoints.
23 Sep	FDI21 campaign	Six bounded attempts diagnosed robot-load, profile, audit identity, evidence capture and runtime cost. RRTConnect completed in r6; RRT/RRT* comparison remained incomplete.
24 Sep	FDI11 operator trial	All three rows stopped at shared PreEntry IK before planner path search. This reframed the problem as feasibility/robot configuration, not planner ranking.
25 Sep	Renovation plan	P0 moved to the engineering workbench: task-independent state checks, J1-J5 jog, Base/Home review, records, two-area UI and strict preview authority.
26-27 Sep	Source closure	Guarded jog correlation, unknown latch/reconciliation, invalid-state review, Base/Home staging, ordered evidence/export/reopen and two-area authority passed focused host suites.
27 Sep	Headed Base/Home	The reviewed case passed the bounded in-app Base/Home happy path; whole-process video remained partial because shutdown exited 1.
28 Sep	Five-DOF refactor	URDF, Movelt, guard, bridge, workflow, persistence and UI now use exactly J1-J5. The spindle remains fixed geometry outside IK.
28 Sep	TCP and lifecycle	No-case TCP drag/IK workbench passed a complete headed recording. The isolated no-case connected shutdown condition also exited cleanly.
28 Sep	Case campaign r13	Passed restore, 31-object scene, workspace, one accepted J1 jog and invalid draft. Stopped at cached Guarded Jog availability before unknown-result injection; source fix passed 79 tests.
28 Sep	Halt	Prepared r14 was blocked before launch by account usage/approval. Operator halted further runtime testing; evidence was preserved.

4A. What was accomplished successfully

Current implementation capability

Capability	Verified implementation state
Five-DOF model	Exact J1-J5 shape enforced across current state, Home, IK, limits, bridge and guard. J6 is excluded rather than silently truncated.
Guarded joint exploration	J1-J5 sliders/numeric controls and keyboard paths stage drafts. Exact acknowledged acceptance advances the simulated robot; rejected/unknown requests preserve evidence.
Detached Base/Home	Stage, review, cancel, accept and uncertainty reconciliation paths keep candidates separate from accepted robot/scene state.
Read-only diagnostics	Draft state can be inspected for mechanical/reviewed limits, native static validity, FK, target residual when available, TCP and drill-axis evidence.
Records and history	Schema 1.0 stores ordered requested/accepted/rejected/unknown events. Export and historical reopen/replay are display-only.
TCP workbench	Explicit-enable goal widget, mouse drag, X/Y/Z and pitch/yaw nudges, five-joint IK staging and disable cleanup passed a no-case headed campaign.
Two-area interface	Planning & Diagnostics and Preview & Control share one authority model. Blocked/partial/manual paths cannot promote a live plan.
Lifecycle	Caller-owned VTK plan/FK results are released exactly once. The isolated no-case connected shutdown run exited 0 without process residue.
Case display	Opened-jaw proxy visibility and separate-copy save/reopen passed a bounded probe; the original case was unchanged.

238 TCP/UI/bridge/facade/shell host checks	70 Full bridge suite	35 Manual Jog UI suite	44 Headed-runner source suite
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5. Multi-planner comparison workflow

Implemented diagnostic capability, incomplete benchmark

Current case identity. Saved paths are display-only evidence, not ROS/Moveit or guard-valid plans. Use a fresh single-planner replan before guarded preview.

	Planner	Result	Full task	First blocker	Waypoints
1	RRTConnect...	Fail	Blocked	Full-chain ...	461
2	RRTkConfig...	NotRun	NotRun		0
3	RRTstarkCon...	NotRun	NotRun		0


```

{
  "message": "Approach ready: 80 free-space, 12 fixed-axis Stage-2 checkpoint(s) from 429 Moveit samples. The independent phase guard keeps all non-tool collision rules strict while suppressing only configured burr-to-task contact; every suppression will be reported. Route selection used only controllable arm joints; the pneumatic spindle is external and not planned. through previously validated workspace clearance sample 198. Full-task status is Blocked at Stage 3; the provisional Approach preview remains available as historical evidence and is not a drilling authorization. WARNING: requested combined insertion 8.071 mm (drilling depth 7.571 mm + provisional guide/shell allowance 0.500 mm) exceeds the provisional combined insertion envelope 6.500 mm by 1.571 mm. Effective tool protrusion is 7.000 mm with 0.500 mm remaining visible/protrusion clearance; no shell measurement or calibration is claimed. The exact approved Entry\u2192Target line will be planned; it was not shortened or moved. Physical fit pass: 1.00 mm burr \u2264 1.50 mm guide bore.",
  "paths": {
    "stage1": [
      {
        "link-1_Revolute-1": 0.0,
        "link-2_Slider-2": 0.0,
        "link-3_Revolute-3": 0.0
      }
    ]
  }
}

```

Replay Saved Path (display only) Use Planner for Next Fresh Replan Close

Figure 1. FDI21 r6 retained RRTConnect row. It records a blocked full task with 461 waypoints and keeps saved paths display-only. SHA-256 aa4591b3...d33bf4. This is diagnostic evidence, not an accepted route.

FDI21 r6 result

- Frozen identity: case, branch, task, trajectory, robot profile, Base, Home and collision audit were checkpointed.
- RRTConnect completed: Approach and Stage 2 passed; Stage 3 stopped at composed waypoint 459 on non-approved target-tooth to pneumatic-spindle contact.
- The requested combined insertion was 8.071 mm, exceeding the provisional 6.500 mm envelope by 1.571 mm.
- RRT began but the 20-minute cap interrupted its second waypoint guard. RRT* did not run.
- No relative planner-speed conclusion is valid. No combined accepted comparison package or preview authority was produced.

Where the time went

Measured component	Observed time	Interpretation
RRTConnect trial	about 874 s	Full diagnostic row including candidates, routes, Cartesian checks and guard.
8 waypoint-guard calls	813.39 s	Dominant cost; repeated full native guard work.
14 joint-plan calls	4.60 s	OMPL plan requests were a small fraction of the row.
14 Cartesian calls	21.97 s	Material but far below waypoint-guard cost.

5A. Planner diagnostic follow-on

What the screenshots taught us - and what they could not prove

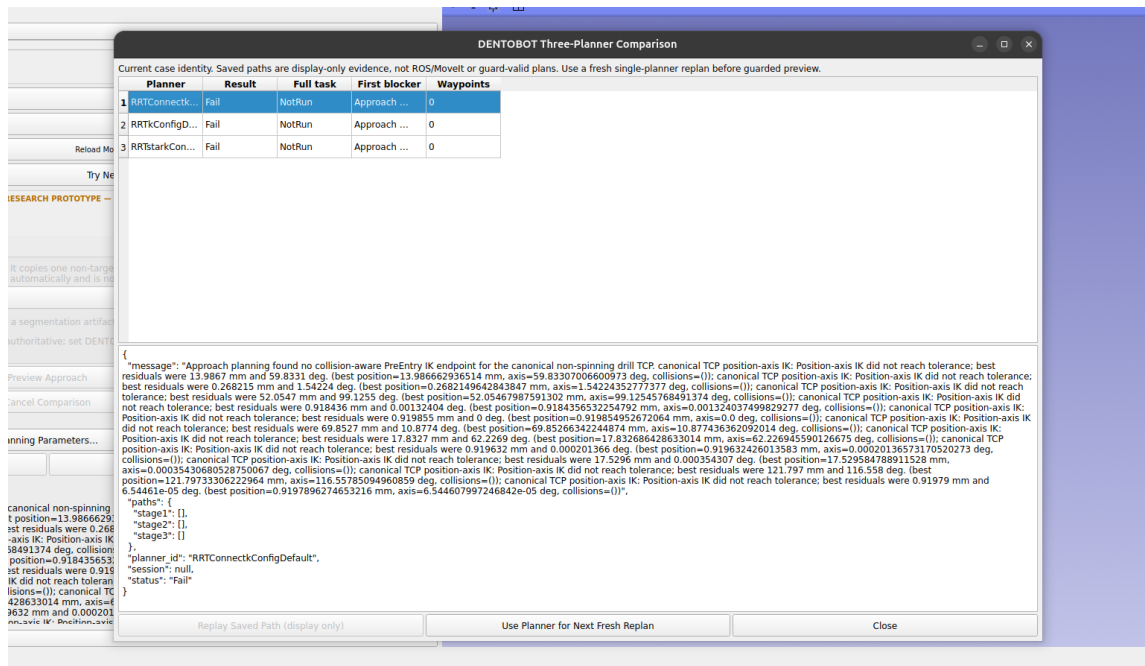


Figure 2. Tarun's FDI11 normal-window comparison. All three configured rows fail at the shared Approach/PreEntry IK prerequisite with zero waypoints and full task NotRun. SHA-256 bedd542c...581e84. The planners were not comparatively exercised.

The earlier FDI21 collision recapture attempted inferior, apical and oblique views from the retained first-invalid state. Tarun rejected them because the rendered tooth and spindle were separated. Source attribution then found the recapture incorrectly applied the lower-jaw opening transform to a fixed upper-jaw target. A fail-closed overlap check now prevents such images from being labelled collision evidence.

Lesson: an image is useful evidence only when its geometry payload, transform, identity and exact evaluated robot state match the native guard record. The invalid recapture remains evidence of an attribution defect, not proof that the guard was wrong.

Together, the FDI21 and FDI11 comparisons supported the renovation decision. FDI21 showed a late physical/scene constraint after earlier stages passed; FDI11 stopped before planner search at five-axis endpoint feasibility. Changing only the OMPL algorithm cannot answer either problem reliably.

6. Diagnostic run summaries

Representative headed evidence from the renovated workflow

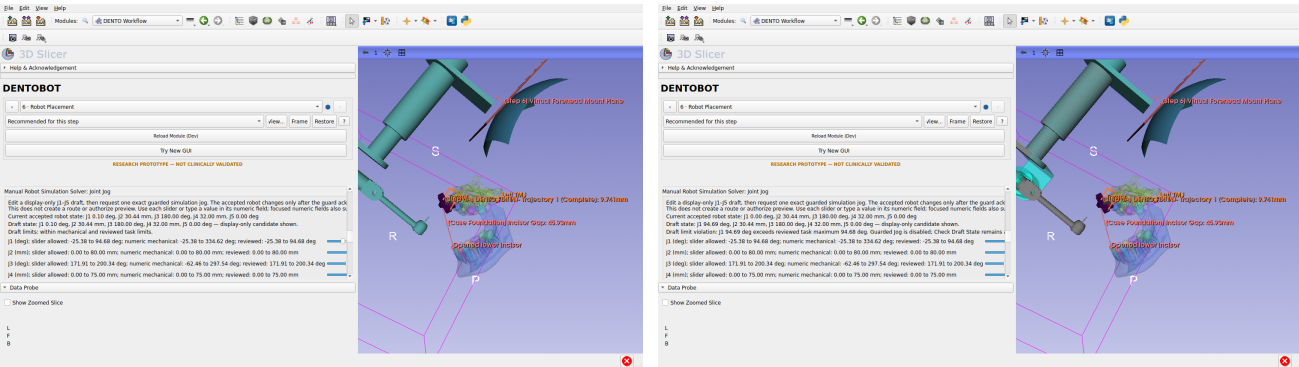


Figure 3. Accepted 0.1-degree J1 guarded simulation jog. Accepted, monitored and displayed state converged. SHA-256 0e1c3c28...e5b86.

Figure 4. Out-of-reviewed-range draft remains visible and blocks Guarded Jog without advancing accepted state. SHA-256 a653d8fe...93ea3.

Run	Truthful result
Base/Home headed r2	Passed bounded case/scene, guarded J1, Base stage/cancel, zero-displacement Base acceptance and current-pose Home acceptance. Video partial because Slicer exited 1.
Five-DOF migration r1-r3	Exposed exact profile mismatch, stale cached parameter node and mutation-before-audit ordering. Each failure was retained; restore ordering was corrected.
Five-DOF campaign r13	Passed strict reopen, ROS and 31-object scene, workspace, accepted jog and invalid draft. Stopped before controlled unknown injection because Guarded Jog availability stayed cached false.
Completion-order fix	Shell now clears busy, completes request and performs an authoritative UI refresh in that order. Combined source checks passed 79 tests.
Prepared r14	Wrapper and diagnostics prepared; command was blocked before launch by account usage/approval. No process, video, screenshot or case save started.

r13 recording SHA-256: 4b70307c9e5a1fd3d68ed651f96d20bbc9c388d724bec99df2e77d7de61f5d09. It is partial failure evidence because the runner exited nonzero, even though the recorder finalized and owned processes were cleaned up.

6A. TCP workbench and lifecycle evidence

Complete bounded no-case campaigns

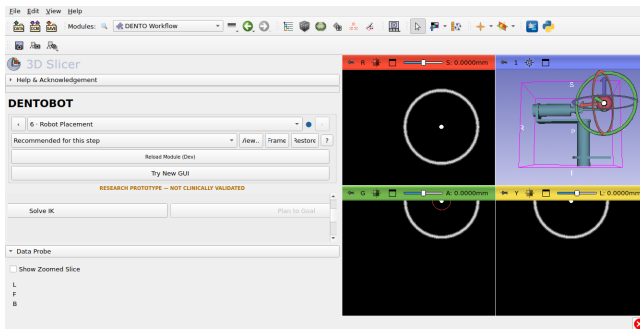


Figure 5. TCP goal widget after orientation nudges and five-joint IK staging. The workbench passed default-off gating, mouse drag, directional/orientation controls and disable cleanup. SHA-256 bab77fd3...d410c.

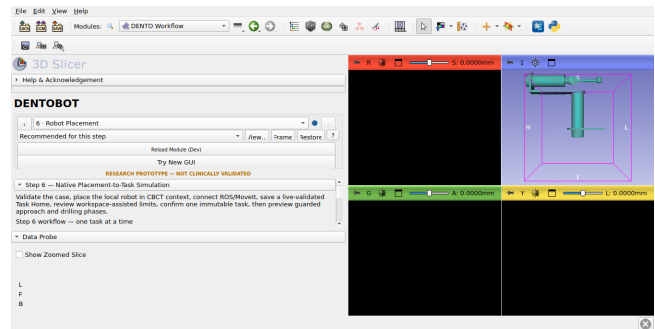


Figure 6. Isolated connected no-case shutdown campaign. Slicer and recorder exited 0 with no class-loader/fatal marker or owned-process residue. SHA-256 e269565b...28bdb.

TCP workbench evidence boundary

- Passed: explicit Enable TCP Drag, live mouse drag, X/Y/Z and pitch/yaw nudges, position-axis J1-J5 Solve IK staging, accepted-state separation and disable cleanup.
- Passed: zero route/preview authority and complete 1600x900 recording with Slicer exit 0.
- Pending: case-bearing collision-aware TCP drag, physical keyboard delivery and measured normal-window responsiveness.

Shutdown evidence boundary

The no-case connected S6-U-01 reproduction is closed. A case-bearing whole-run shutdown remains part of the final campaign and cannot be inferred from this narrower pass.

7. Stable, verified and presentable now

What can be demonstrated at the weekly meeting without overstating acceptance

Presentable area	Stable claim
Architecture	Five-DOF arm model and workflow contract; J6 removed from IK/state/Home/guard; spindle retained as separate fixed geometry.
Engineer workbench	Joint draft controls, exact guarded acceptance, invalid-state retention, Base/Home review and uncertainty model.
Cartesian exploration	Explicit-enable TCP drag and directional/orientation controls with five-joint IK staging in the bounded no-case headed run.
Evidence	Ordered schema-1.0 records, JSON/readable export, display-only historical reopen/replay and state-matched screenshots.
Safety/authority	Manual, partial and historical paths never authorize a route or preview. Fresh complete preflight identity is required.
Diagnostics	Three-planner UI and evidence pipeline, FDI21 runtime-cost attribution, FDI11 shared IK prerequisite finding and strict limits on conclusions.
Lifecycle	Exact VTK result ownership release and isolated no-case clean shutdown.

DEMO READY	No-case TCP workbench recording and screenshot sequence.
EVIDENCE READY	Five-DOF r13 accepted jog and invalid draft; FDI21/FDI11 planner diagnostics with explicit limitations.
SOURCE READY	Guarded state model, records, Base/Home review, two-area UI and preview authority boundaries.
PENDING	Representative case-bearing whole-run campaign and Tarun's normal-window robotics/usability verdict.

Recommended meeting wording: 'The workbench is implemented and several bounded simulation slices are verified. Final representative case acceptance and operator usability review remain open.'

8. What failed or remains unproved

Failures were retained as diagnostic evidence, not hidden by retries

Area	Status	Current evidence
Three-planner benchmark	Incomplete	RRTConnect completed; RRT was interrupted; RRT* did not run. No fair ranking.
Collision recapture	Invalid evidence	Wrong jaw-transform attribution displaced the upper target. Images cannot support geometry advice.
Case restore	Three diagnosed failures	Profile migration, cached parameter node and audit ordering were corrected; strict validation remains.
Case display/save	Bounded pass only	Separate-copy reopen passed; Tarun's visual verdict remains pending.
r13 campaign	Partial	Stopped before rejected/unknown/reconciliation because availability was cached while busy.
r14 campaign	Not launched	Account usage/approval rejected the runtime command before any process started.
Whole-process shutdown	Case-bearing open	No-case connected run exited cleanly; prior case runs exited 1 after functional evidence.
Full-chain preview	Not exercised	Current fixture lacked a strict-current confirmed task/complete route.
Responsiveness	Not measured in operator window	Host/source counts and progress callbacks do not prove normal-window interaction latency.

Non-negotiable evidence boundaries

- No hardware motion, drilling or patient-facing operation was performed or authorized.
- Host tests are not headed GUI acceptance; headed no-case tests are not case-bearing acceptance.
- A configured planner ID is not by itself proof of which algorithm executed.
- A screenshot is not native collision proof unless exact geometry, transform and evaluated state match.
- Tarun's normal-window verdict remains PENDING and must not be inferred from automation.

9. Next week plan

Resume only after explicit runtime instruction; finish Step 6 before integration work

Order	Gate	Expected acceptance evidence
1	Source review freeze	Review the current dirty diff, keep five-DOF/J6 exclusion and authority boundaries intact, and preserve all existing evidence.
2	Representative case	Create or select a newly reviewed five-DOF .dentocase with exact Base, validated Home, confirmed task and current resource fingerprints.
3	Narrow runtime completion	Verify native accepted, rejected, stale and unknown guarded J1-J5 outcomes plus read-only reconciliation on that case.
4	Base/Home uncertainty	Exercise unknown commit reconciliation without a second publish/save and prove accepted state changes only after authoritative confirmation.
5	Case-bearing TCP	Enable TCP drag explicitly, stage collision-aware IK review, check keyboard delivery and confirm no preview authority.
6	Record lifecycle	Export ordered motion evidence, reopen display-only history in a fresh process, and prove no live-state restoration.
7	Full-chain authority	With an eligible task, run Home-to-PreEntry-to-Entry-to-Target preflight, positive preview, Stop/interruption and guarded Return Home.
8	Responsiveness	Measure interaction and rendering in the normal window; separate GUI/render/FK latency from guard and ROS call duration.
9	One final campaign	Record the complete headed GUI checklist with JSON, screenshots, logs, video, hashes and exact process cleanup; stop at the first causal failure.
10	Operator verdict	Tarun reviews robotics meaning, invalid-state usefulness, TCP controls, visual clarity and workflow usability.

Explicitly downstream

Only after Step 6 is complete: curate proven 5.10 integrity/performance changes from the parallel checkout into a fresh reviewed integration worktree. Do not merge the mixed 5.12 branch wholesale. Case-specific planner comparison and automated Base-pose/design sweeps remain later work unless the operator changes priority.

Expected next-week outcome: one current five-DOF case, all manual uncertainty gates exercised, and one complete case-bearing recording ready for Tarun's normal-window verdict.

10. Verification ledger

Recorded checks and what they establish

Check	Result	Meaning
Five-DOF indexed host selection	394 passed	Source/build/runtime contract selection; not full GUI campaign.
TCP/UI/bridge/facade/shell	238 passed	Workbench source integration.
Manual Jog UI	35 passed	Draft, availability, completion ordering and reconciliation UI rules.
Headed runner source	44 passed	Automation/checklist source behavior.
Combined latest	79 passed	Manual-jog plus headed-runner source regression.
ROS2 bridge full suite	70 passed	Bridge conversions and ownership/release paths.
VTK release focus	8 passed	Exactly-once release in normal, malformed and exception paths.
Earlier renovation gates	181 / 225 / 251 passed	Historical source and reviewed-integration checkpoints.
TCP no-case headed	PASS, exit 0	Explicit-enable drag/IK controls and clean owned-process exit.
S6-U-01 no-case headed	PASS, exit 0	Isolated connected teardown condition closed.
r13 case campaign	PARTIAL	Accepted jog and invalid draft passed; unknown scenario not reached.

Current provenance

Field	Value
Checkout	/home/light-tarun/dentobot/ros2_ws/src/DentoBot-step6-renovation
Branch	feature/step6-workflow-renovation-20260925
HEAD	84234a684f026498a5bc28a425b97d13646e7001
Source case	/home/light-tarun/dentobot/data/Slicer_Saved/SampleStudy1/SEPT24/sept28_fdi11_step6.dentocase
Source case SHA-256	e21fee7f850aa3517ce26e7920bf7dde7c06d3dda7dd6de7e360e7fc69a32786
Expected new case	sept28_fdi11_step6_five_dof_acceptance.dentocase - absent

11. Evidence index

Primary diagnostics, recordings and screenshot hashes

Evidence	Path	Contents
FDI21 planner r6	data/dentobot-runs/step65-three-planner-fdi21-31-20260923/fdi21-r6/	checkpoint.json, timeline.jsonl, stack.log, Base/Home/row screenshots
FDI11 comparison	Workspace/docs/diagnostics/evidence/fdi11-step6-2026-09-24/	Eight operator screenshots and saved-pose audit
Base/Home acceptance	data/dentobot-runs/s6-live-01-base-home-accept-20260927T131832Z-r2/	Run diagnostics, checklist, screenshots and partial video
Five-DOF r13	data/dentobot-runs/s6-live-01-five-dof-campaign-20260928T164300Z-r13/	Itemized JSON, state-matched screenshots, logs and partial MKV
Prepared r14	data/dentobot-runs/s6-live-01-five-dof-campaign-20260928T165722Z-r14/	Pre-launch wrapper and diagnostics; no runtime output
TCP workbench	data/dentobot-runs/step6-tcp-workbench-headed-20260928-9/	Five screenshots, JSON, complete video and cleanup
Clean shutdown	data/dentobot-runs/s6-u01-final-headed-20260928-0hCREB/	Connected screenshot, complete video, cleanup and diagnostics
Display/save/reopen	data/dentobot-runs/step6-view-u01-headed-ugfni53t/	Open/reopen screenshots and separate-copy evidence

Artifact	SHA-256
FDI21 RRTConnect diagnostic PNG	aa4591b3f09093cfaeeca22e90ec2ce2fc9604e2ff160b361aed82d678d33bf4
FDI11 later comparison PNG	bedd542c080c1f0c9389535b42721f55b64dfb43ba87ecbe632f70f621581e84
TCP workbench orientation/IK PNG	bab77fd3d07e03870cca007028e7f75dbc2e2399ff289098b36da6e4dcd410c
r13 accepted jog PNG	0e1c3c28fd6de63623bc3197faace47aa12ec0a907adcc4482e9044c78e5b86d
r13 invalid draft PNG	a653d8fe2a0b70f038fde1efa18398d1ca37fb566a421d52d5e79ee862693ea3
No-case shutdown connected PNG	e269565bbe208a3c375b2db970a1aaf21ff1503136c42407ec60ef5eee428bdb

Canonical records

- Workspace/docs/diagnostics/STEP6_RENOVATION_IMPLEMENTATION_PLAN_2026-09-25.md
- Workspace/docs/diagnostics/STEP6_FIVE_DOF_DETAILED_ACCEPTANCE_CHECKLIST_2026-09-28.md
- Workspace/docs/diagnostics/FDI31_GUI_PLANNER_P0_PLAN_2026-09-21.md
- Workspace/docs/logbook/2026-09-23.md through 2026-09-28.md
- Workspace/docs/backlog.md, TASKS.md, DEVELOPMENT_PLAN.md and DECISIONS.md

12. Closing assessment

Development is on track, with a clearly bounded acceptance gap

The week produced substantial engineering progress. It converted Step 6 from a mostly planner-facing sequence into a five-axis robotics workbench with explicit state ownership, inspectable failure modes, guarded manual movement, Cartesian probing, historical evidence and strict preview authority. It also turned the expensive multi-planner campaign into useful performance and geometry diagnostics instead of treating an incomplete comparison as a planner verdict.

The project is not finished. The present source is ahead of its representative runtime evidence. The key remaining task is one current case that exercises native rejected/unknown/reconciliation behavior, case-bearing TCP review, complete route authority, interruption and fresh-process record reopening under a single fully recorded campaign. Tarun's robotics/usability review then provides the acceptance that automation cannot supply.

Presentable now: architecture, diagnostics, five-DOF state model, manual workbench, no-case TCP campaign, bounded accepted/invalid jog evidence, planner failure analysis and clean no-case shutdown.

Not yet presentable as complete: a representative case-bearing whole workflow, relative planner benchmark, full-chain preview/Stop/Return campaign, measured operator-window responsiveness and final operator verdict.

Prepared from development-controlled records and retained diagnostic evidence.

No engineer-owned Daily Compass, Personal Work Journal or Project Tracker artifact was read or modified for this report.